

Forensic “lenses” — diagrammatic summary

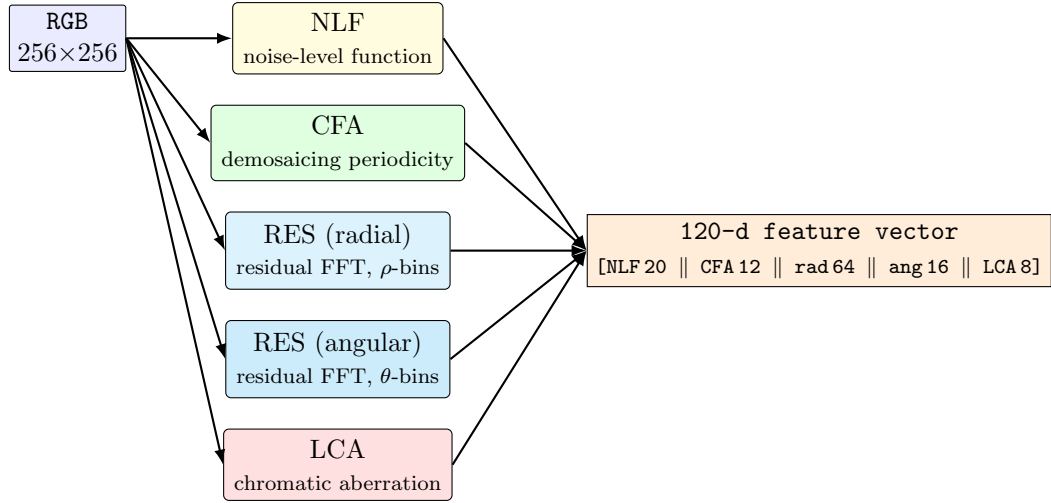
ADAIS / wave_1

Source. `SRC/lens_features_extractor.py`. Four pure-numpy feature families that target *positive camera provenance* (capture-time physical traces) rather than absence of generative artefacts.

Block budget (default config).

Block	dim	contents
NLF	20	16 curve bins + {a, b, R^2 , valid_frac}
CFA	12	3 channels $\times$ {peak_hh, vh, dh, pool}
RES, radial spectrum	64	log-spaced radial bins, L1-normalised
RES, angular spectrum	16	sectors of 11.25° in $[0, \pi)$
LCA	8	2 pairs (R $\rightarrow$ G, B $\rightarrow$ G) $\times$ {slope, R^2 , mean_tan, std_rad}
total	120	

1. Pipeline overview



Each block is independent; image is loaded once, the shrink-age residual is computed once and shared by the two RES blocks.

Figure 1: Top-level layout. The image enters once; five lenses run in parallel and their outputs are concatenated.

2. NLF — Noise-Level Function

Idea. A real sensor produces *heteroscedastic* noise $\sigma^2(I) = aI + b$ (Poisson–Gaussian). AR/diffusion decoders do not.

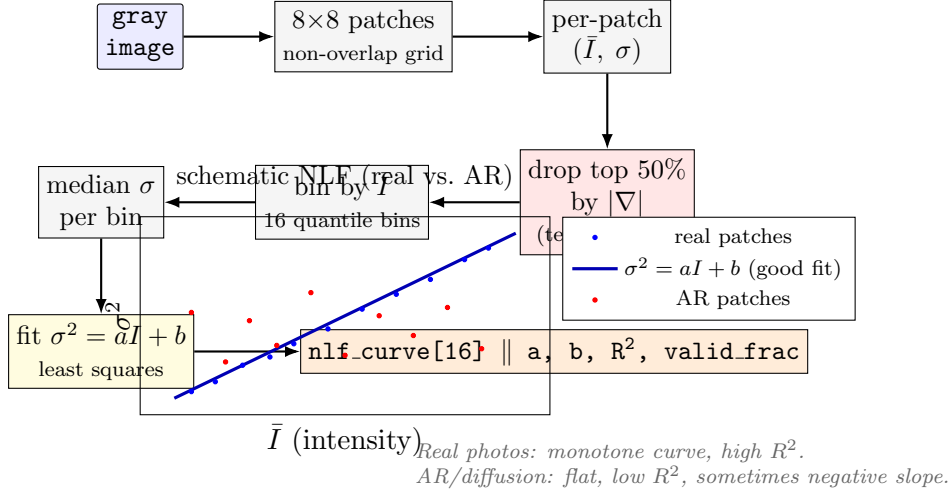


Figure 2: NLF block. Patches are filtered for content, binned by mean intensity, and the resulting $(\bar{I}, \sigma)$ curve is fit to the Poisson–Gaussian model. The fit residuals are themselves informative.

3. CFA — Colour Filter Array periodicity

Idea. Bayer demosaicing leaves a 2-pixel periodic correlation in each channel’s high-pass residual — visible as peaks at the half-Nyquist locations of the 2-D FFT.

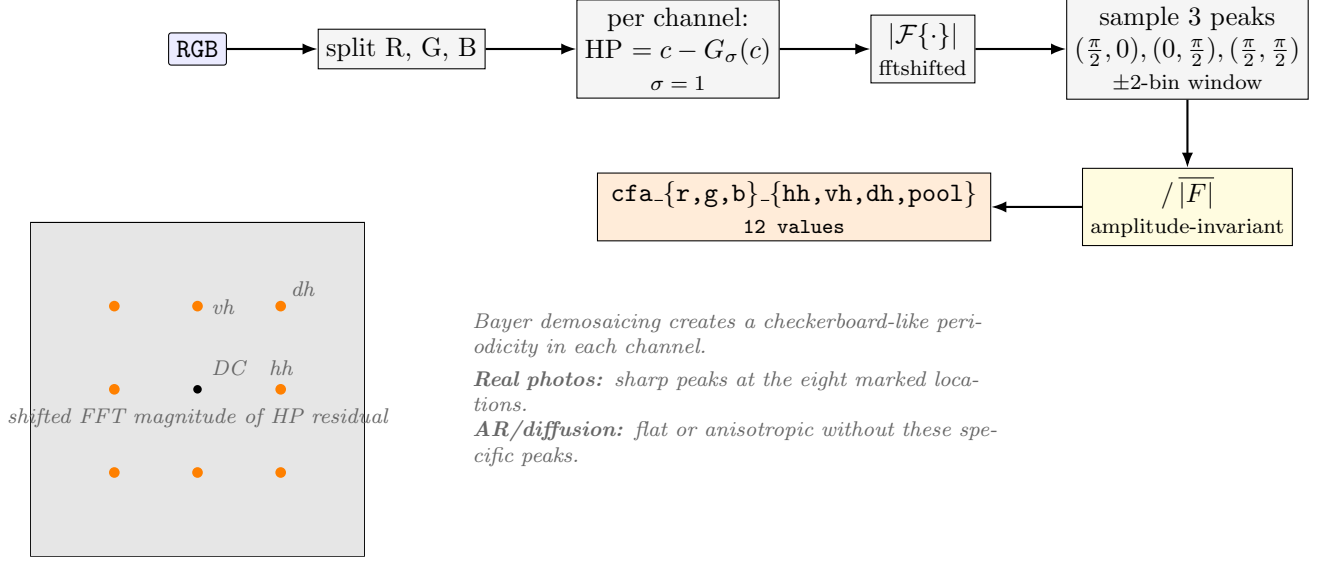


Figure 3: CFA block. Each of $\{R, G, B\}$ is high-passed, FFT-magnitudes are sampled at the three half-Nyquist diagnostic frequencies, and normalised by the mean spectrum to make the score robust to overall residual energy.

4. RES — residual spectrum (radial & angular)

Idea. The shrinkage residual (wavelet, by default) of a real photo has a smooth, monotone fall-off in frequency. Decoder up-sampling kernels leave periodic replicas: **radial bins** surface their presence, **angular bins** surface their orientation (each generator’s up-sampler has a different signature).

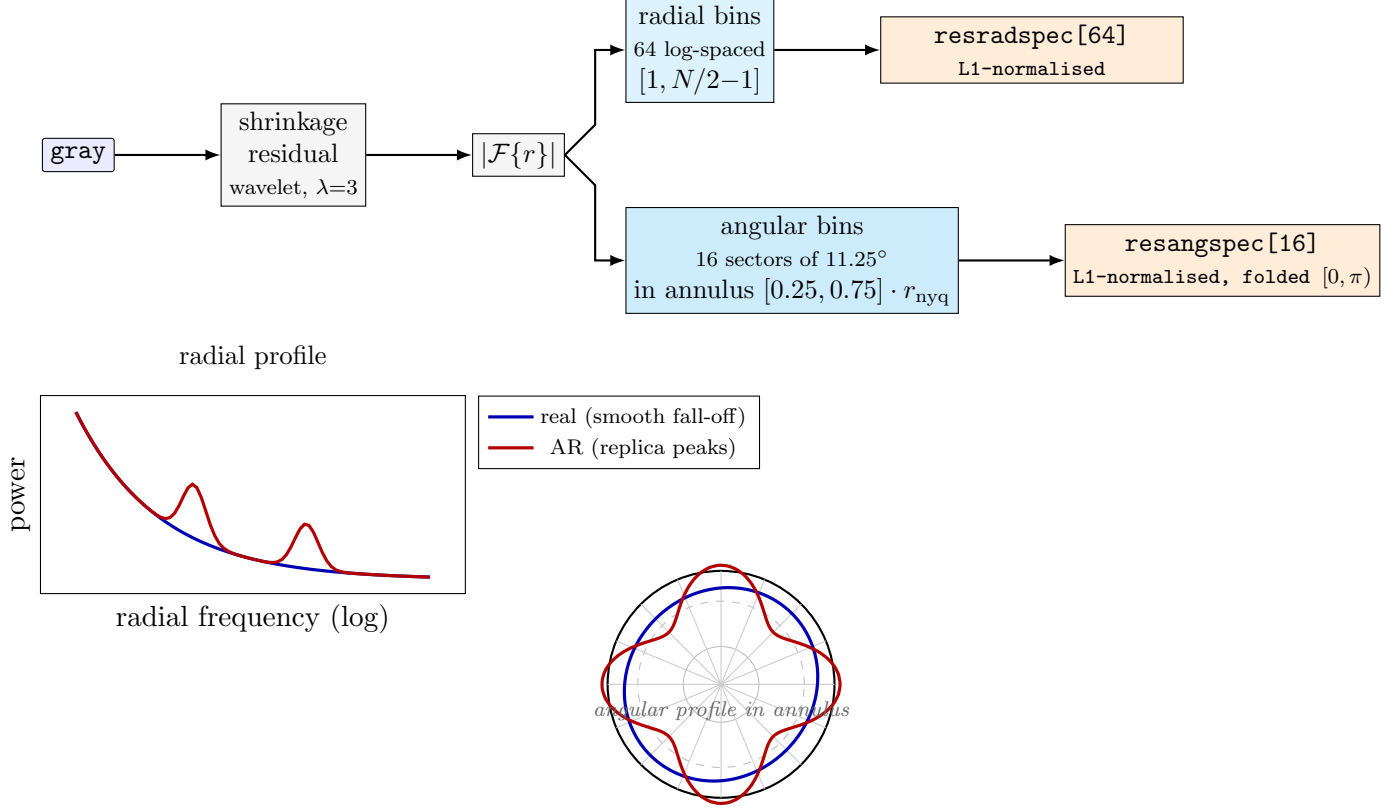


Figure 4: RES block. The same FFT magnitude is binned two ways: radial bins (top) on a log scale cover the full Nyquist range and discriminate *real vs. synthetic*; angular bins (bottom) live in a mid-frequency annulus and discriminate *synthetic vs. synthetic* via the orientation pattern of each decoder’s up-sampler.

5. LCA — Lateral Chromatic Aberration

Idea. A real lens refracts wavelengths slightly differently, so the R and B channels are shifted radially outward from the optical centre. The shift grows linearly with distance from the centre. AR generators have no such physics. LCA generators have no such physics.

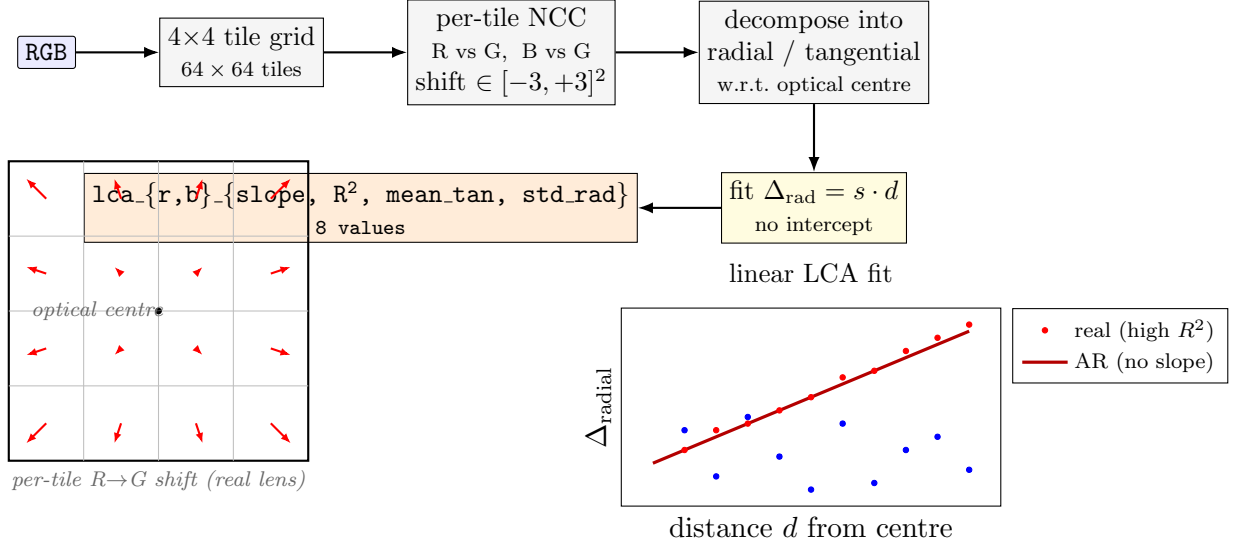


Figure 5: LCA block. R and B channels are aligned to G on a 4×4 tile grid via integer NCC search; per-tile shifts are decomposed into a radial component (along the line to the image centre) and a tangential component. A real lens gives radial shifts that grow linearly with distance and a near-zero tangential mean.